

HEProAI Activity 2: Student Scoring System Logic

Dedicated Mentoring System for Students

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1. Executive Summary

This document details the rule-based intelligence applied to transform raw data from `students.csv` into actionable performance metrics. By normalizing variables like GPA (0–10) and Attendance (0–100%), we establish a standardized **Student Readiness Index (SRI)**. This allows mentors to identify student needs across four critical behavioral domains.

2. Domain Scoring Formulas

Each domain score is normalized to a scale of **0–100**.

2.1 Academic Performance Score (APS)

Reflects classroom performance and consistency.

$$APS = \frac{(GPA \times 10) + Attendance}{2} \quad (1)$$

2.2 Wellness & Wellbeing Score (WWS)

Prioritizes mental health. Stress is treated as a negative indicator.

$$WWS = \frac{(10 - Stress) \times 10 + MentalWellbeing \times 10 + SleepHours \times 10}{3} \quad (2)$$

2.3 Productivity & Time Management Score (PTMS)

Measures workload efficiency and completion rates.

$$PTMS = \frac{AssignmentCompletion + (ProductivityScore \times 10)}{2} \quad (3)$$

2.4 Career Readiness Score (CRS)

Evaluates alignment with professional goals and job market skills.

$$CRS = \frac{(CareerClarity \times 10) + (SkillReadiness \times 10)}{2} \quad (4)$$

3. Student Readiness Index (SRI)

The **SRI** represents the holistic state of a student, calculated as the arithmetic mean of the four domain scores.

$$SRI = \frac{APS + WWS + PTMS + CRS}{4} \quad (5)$$

4. Classification Thresholds & Action Logic

Students are classified into color-coded categories based on their SRI to guide intervention strategies.

Category	SRI Range	State	Recommended Action
Blue	85 – 100	Excellent	Leadership & Peer Mentoring
Green	70 – 84	Stable	Routine monthly check-ins
Yellow	50 – 69	Needs Attention	Targeted domain support (e.g., WWS)
Red	less than 50	At Risk	Immediate human intervention

5. Validation Results

The logic was validated using behavioral patterns from Activity 1:

- **Burnout (S1001-S1005):** Low WWS and PTMS results in a **Red/Yellow** SRI.
- **Hidden Struggle:** Low GPA with high attendance is flagged via APS.
- **Career Uncertainty:** High academic success with low CRS triggers career playbooks.